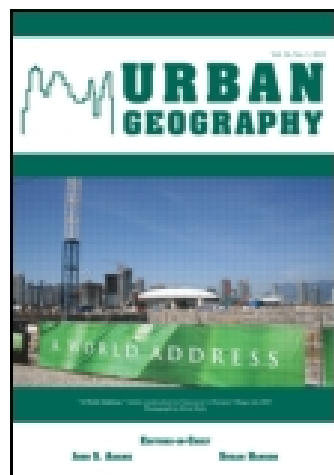




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### Industrial Land Development and Manufacturing Deconcentration in Greater Jakarta

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## INDUSTRIAL LAND DEVELOPMENT AND MANUFACTURING DECONCENTRATION IN GREATER JAKARTA

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*Abstract:* Industrial land development has become a key feature of urbanization in Greater Jakarta, one of the largest metropolitan areas in Southeast Asia. Following Suharto's market-oriented policy measures in the late 1980s, private developers have dominated the land development projects in Greater Jakarta. The article investigates the extent to which these private industrial centers have effectively reduced the domination of Jakarta in shaping the entire metropolitan structure. The analysis indicates that major suburban industrial centers have captured most of the manufacturing employment that has dispersed from Jakarta. The industrial centers have now increasingly specialized and diversified. It is likely that a polycentric metropolitan structure will emerge in the future. [Key words: deconcentration, industrial park, manufacturing, suburbanization, Jakarta.]

### INTRODUCTION

Jakarta is the most suburbanized mega-city in South East Asia (Murakami et al., 2003). Since the 1990s, the population of Jakarta's suburbs has surpassed that of its metropolitan core (Hudalah and Firman, 2012). In relation to this population dispersal, other important

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elements of Jakarta's suburbanization have entailed massive urban land-use conversion and large-scale housing development (Firman, 2004; Winarso and Firman, 2002). The land subdivision projects have mainly taken the form of dormitory and industrial towns. In the peak period of the 1990s property boom, more than 23 new towns ranging from 500 to 6,000 hectares were developed around Jakarta alone (Firman, 2004; Winarso and Firman, 2002).

While population deconcentration is well documented, there has been little attention paid to the role of employment restructuring in shaping this suburbanization trend in Indonesia. North American and European literature on the subject has principally focused on deconcentration of the service sector, seen as the economic engine of post-industrial cities (Boiteux-Orain and Guillain, 2004; Coffey and Shearmur, 2002; Gilli, 2009; Lang et al., 2009; Shearmur and Alvergne, 2002). In contrast, it is argued that in industrializing East Asian countries, such as in China, it is the manufacturing sector that is driving changes in the spatial structure of major metropolitan areas (Feng and Zhou, 2005; Feng et al., 2008; Wu and Phelps, 2008). In Indonesia, an earlier study conducted in the 1990s concluded that Jakarta "remains a major corporate manufacturing center" in the metropolitan area (Henderson and Kuncoro, 1996, p. 93). However, it is important to note that since then the planned agglomeration of industries, in the form of industrial parks and town projects, has grown rapidly in the suburbs to anticipate and accompany massive inflows of FDI (Foreign Direct Investment). Therefore, updated studies need to be conducted to uncover new relationships that are possibly emerging between the metropolitan core and the suburbs as a result of this industrial restructuring process.

This article describes the current trend of economic suburbanization in Greater Jakarta. It first assesses the extent to which manufacturing industries have deconcentrated in Indonesia's largest metropolitan area. It also explores the effect of this industrial deconcentration on the metropolitan area's spatial structure.

After reviewing the current literature on employment deconcentration, the article documents the trend toward industrial suburbanization in Jakarta over the past two decades. It also reviews some of the socio-political and economic factors contributing to this trend. Our empirical analysis describes the spatial structure produced by this deconcentration. It is argued that industrial suburbanization has expanded the boundaries and reshaped the once sprawling, monocentric metropolitan area so that it is now more polycentric with a number of specialized subcenters.

## DECONCENTRATION

Deconcentration can be defined as the process through which a metropolitan area—still recognizable as a city—evolves into an urbanized region. It is often revealed by a massive dispersal of population and employment toward the suburbs (Guillain et al., 2006). Other indicators of deconcentration have focused on the decline of the CBD (the Central Business District) or metropolitan core vis-à-vis its suburbs (Shearmur and Coffey, 2002). Broadly speaking, therefore, a general definition of a deconcentrating metropolitan area could be a region in which there is a decline in population and employment share of the metropolitan core and a corresponding rise in the share of its suburbs (Carlinio and Chatterjee, 2002).

Studies of intra-metropolitan deconcentration can be divided into two often interrelated themes. The first theme is population deconcentration (Bontje, 2001; Dura-Guimera, 2003), which in many cases also includes retailing (Feng et al., 2009; Lang, 2003). This article is focused on the second theme: employment deconcentration or decentralization of jobs. In the context of post-industrial society in developed countries, analyses of employment deconcentration have recently tended to focus on the business service sector, notably, office suburbanization (Boiteux-Orain and Guillain, 2004; Coffey and Shearmur, 2002; Gilli, 2009; Lang et al., 2009; Shearmur and Alvergne, 2002): yet some empirical evidence shows that the manufacturing sector can still be an important feature of employment deconcentration in developed countries (Coffey and Shearmur, 2001; hUallacháin and Leslie, 2009; Lewis, 2001).

Whatever its current role, the manufacturing sector played a major part in the changing geographies of employment of European and North American cities during the 19<sup>th</sup> and 20<sup>th</sup> centuries (Harris, 1994; Lewis, 2001; Phelps and Ozawa, 2003). Currently, the manufacturing sector is playing a similar role in the global cities of East Asia, notably China (Feng and Zhou, 2005; Feng et al., 2008; Wu and Phelps, 2008). This shift is not only market driven: it is argued that industrial deconcentration has become part of wider economic development strategies in a number of Asia's developing countries as they experience structural shifts in their national economies toward industrial and service sector activities (McGee, 2007).

In addition to the greater role of changes in manufacturing location, current East Asian employment deconcentration can also be distinguished from its Western counterpart on the basis of underlying process and triggering factors. The North American literature has traditionally emphasized a local market-driven process of deconcentration. The private location decisions are often seen as the main factor behind land development in the suburbs (see, e.g., Garreau, 1991; Kling et al., 1991). Historically, manufacturing establishments in North America have been attracted by the availability of cheaper land in suburban areas (Harris, 1994). Politically, the creation of suburban industrial complexes was largely influenced by endogenous market forces, including changing behavior of, and economic alliances between, developers and local leaders (Walker and Lewis, 2001).

While the Western literature has paid attention to the endogenous, localized, and evolutionary processes of industrial deconcentration, East Asia has a particular interest in the role of dynamic, external and global factors, such as FDI relocation, in transforming suburban space (see, e.g., Dick and Rimmer, 1998; Hudalah et al., 2007). Industrial land development in suburban China is often planned and carried out at a large scale, and integrated into the national development planning policy. As a consequence the national state, rather than municipal or local government, often plays an active role in suburban spatial restructuring (Feng et al., 2008; Wu and Phelps, 2008). However, in Indonesia we will show that whereas the state provides broad guidance (e.g., with regards FDI), it is private land developers that drive the nature and geography of suburbanization.

### *Implications for Metropolitan Spatial Structure and Pattern*

There are discussions among scholars regarding the implications of employment deconcentration on metropolitan spatial structure. Two contrasting perspectives have

been influential in these debates (Garcia-López and Muñiz, 2010; Lang, 2003; Shearmur et al., 2007). First, the centrist perspective argues that, because agglomeration economies and face-to-face contacts remain key factors in location decisions, firms tend to locate close to one another. As a result, employment—even if it decentralizes—re-agglomerates in a relatively limited number of suburban centers, creating a polycentric spatial structure at an urban-regional scale (Anas et al., 1998; Bogart and Ferry, 1999; Gottdiener and Kephart, 1991; Kloosterman and Musterd, 2001). These suburban centers to a great extent replicate the cohesiveness of the old service city, as manifested in, for instance, the emergence of “new downtowns” (Baerwald, 1978) and “edge city” formation (Bontje and Burdack, 2005; Garreau, 1991). A suburban center may also emerge in the form of “technopole”, referring to a (usually locally) planned concentration dominated by high-tech industrial activities (Castells and Hall, 1994).

Meanwhile, the decentrist perspective emphasizes that automobile technology, the telecommunication revolution, and globalization have increased individual mobility and, thus, have fundamentally changed the location decisions of businesses (Friedmann and Miller, 1965; Muller, 1997). There are no longer any distinct advantages for firms to be located within close proximity of each other. Accordingly, suburban employment disperses, scatters randomly, thus creating a chaotic structure at an urban-regional scale (Shearmur et al., 2007). To the extent that employment clusters, it does so in numerous smaller pockets. Unlike edge cities, these employment pockets follow low-density patterns and, thus, have no clear borders (Gordon and Richardson, 1996; Lang and LeFurgy, 2003).

The deconcentration of East Asian metropolitan employment can lead to structures and patterns more complex than the above centrist-decentrist debate implies. In the 1980s, many scholars were concerned with the uncontrolled transformation of the Asian rural-urban interface due to fast-growing extended metropolitan regions. McGee (1991) has termed the spatial form resulting from this process “desakota”, referring to unplanned and mixed land use along highway corridors between large cities. More recently, an expanding middle class and increasing flows of global capital into Southeast Asia have led to the emergence of particular suburban spatial patterns, especially in the form of privately planned and governed residential or industrial towns (Dick and Rimmer, 1998; Firman, 2004; Hudalah and Firman, 2012; Winarso and Firman, 2002). It has been argued that, to a certain extent, the relocation of residential and industrial activities from a Southeast Asian central city to its peripheral areas may indicate a shifting trend from public- to private-controlled space (Shatkin, 2008). Meanwhile, in China, FDI inflows are planned by government and clustered in special administrative areas called Special Economic Zone (SEZ) (Ding and Zhao, 2011). The remainder of this article seeks to investigate the extent to which private industrial centers, which characterize suburbanization outside China, have effectively reduced the domination of the central city of Jakarta in shaping the entire metropolitan structure.

## DATA AND METHODS

The following analysis is essentially descriptive and exploratory in nature. As such it marshals a variety of evidence in order to contextualize the evolving urban

structure of Jakarta's space economy. It examines causal processes and, in particular, the role played by private industrial centers, without pretending to verify the processes empirically. Rather, it aims at describing and, to some extent, corroborating these processes, whilst describing the way in which the space economy has evolved over the last 15 years.

A variety of secondary data was collected in order to conduct this study: these include population and industrial censuses (Central Bureau of Statistics, 1995–2010), land and property market surveys conducted by a global property consultant (1990s–2010s), policy documents, and previous studies. Following this data collection, content analysis of policy documents and previous studies was conducted to explain factors and actors triggering deconcentration.

To describe the spatial distribution of manufacturing employment in Greater Jakarta, the employment-population ratio (E/P) is used. The method has frequently been used to identify suburban centers in deconcentrated metropolitan areas in Europe and the United States (Bourdeau-Lepage and Huriot, 2005; Forstall and Greene, 1997; Guillain et al., 2006; Shearmur and Coffey, 2002). The use of relative density indicators such as the E/P ratio can reduce bias due to the uneven distribution of vacant space and the use of arbitrary (often administratively based) spatial units. The E/P ratio is good at capturing the spatial dynamics of suburbs, in which absolute densities are low but which can play a significant role in providing jobs for other parts of a metropolitan area.

The employment data used in our analysis are from censuses of formal industrial enterprises conducted by the National Statistical Bureau in 1995, 2001, and 2010 (Badan Pusat Statistik, 1995, 2001, 2010). These censuses focus on medium and large-scale industrial enterprises employing over 20 workers. Suburban *kecamatan* or subdistricts and *kotamadya* or municipalities in the Capital Special Province of Jakarta (the CBD/the metropolitan core) were treated as the basic unit of analysis for calculating E/P. Given changes in the subdistrict boundaries between 1995 and 2010, Greater Jakarta has disaggregated into 145 zones, consisting of 140 suburban subdistricts and five municipalities in the metropolitan core.

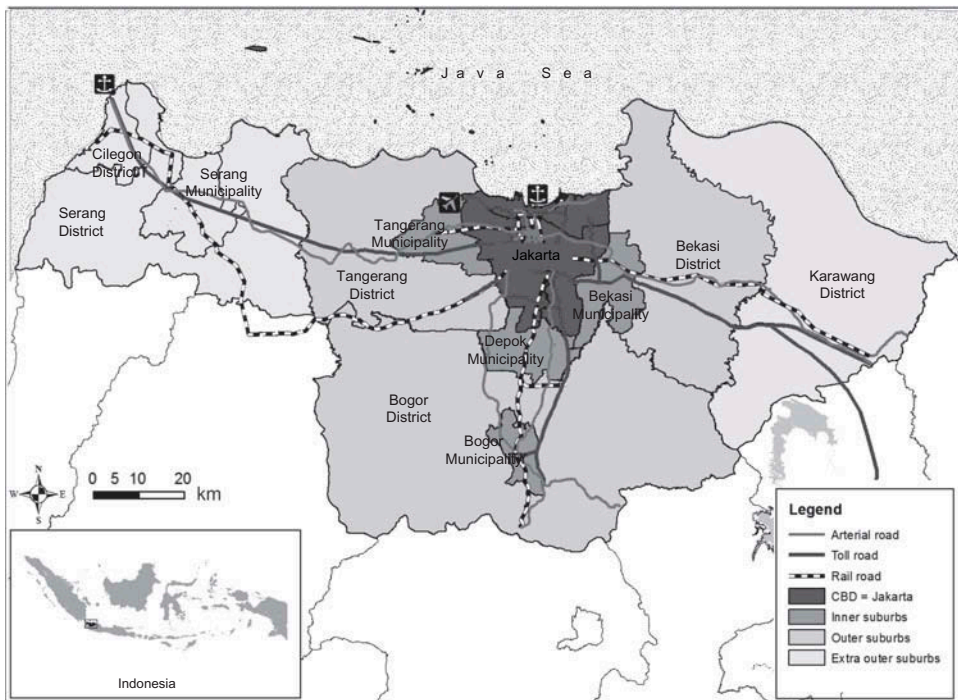
Maps of the evolving E/P indicator are used to illustrate the shifting location of industrial centers in Greater Jakarta. The industrial structures of the centers identified are then compared using Location Quotient (LQ). LQs have often been used to identify concentrations of employment in particular sectors within a metropolitan area (Bogart and Ferry, 1999; Bourdeau-Lepage and Huriot, 2005; Guillain et al., 2006). In this article, LQs represent the share of one sector in an industrial center divided by the share of that sector for all of the analyzed industrial centers. A higher LQ for one sector in a particular employment center denotes a higher likelihood that the industrial center is specialized in that sector.

However, an exception needs to be made for high LQs in centers with relatively low employment: in such cases, even high LQ values can sometimes denote rather small job numbers (Guillain et al., 2006). In our analysis, results for Cilegon should be treated with some caution, since it has only 5,000 to 10,000 jobs (in 1995 and 2010 respectively). These figures are much lower than the average number of jobs—56,000 to 91,000—in industrial centers during the years 1995–2010.

## FROM JABODETABEK TO GREATER JAKARTA

Large-scale private land development has become a major feature of Jakarta's suburbanization. Past studies conducted on Jakarta's suburbanization generally restricted their scope to land development lying within the formal boundaries of the Jakarta Metropolitan Area (JMA), also known as Jabodetabek (Jakarta-Bogor-Depok-Tangerang-Bekasi) (Firman, 2004; Firman and Dharmapatri, 1995; Henderson and Kuncoro, 1996; Hudalah and Firman, 2012; Winarso and Firman, 2002). Jabodetabek consists of Jakarta as the metropolitan core, surrounded by its suburbs including *kotas* or municipalities (the *inner* suburbs) and *kabupatens* or districts (the *outer* suburbs) adjacent to Jakarta, which are also called Bodetabek. Jabodetabek has a total area of 5,898 km<sup>2</sup>.

The development of private industrial parks in suburban Jakarta has followed, or at least has corresponded to the development of, inter-city highways (Henderson and Kuncoro, 1996). There have been at least three major highways constructed between Jakarta and the suburbs. The first highway is Jagorawi (Jakarta-Bogor-Ciawi) Toll Road, which began operating in 1978. It stretches from the city to the southern suburbs. Major industrial zones (*zona industri*) or unplanned industrial areas were developed following the construction of this 60 km highway.



**Fig. 1.** Map of Greater Jakarta. CBD = Jakarta (Capital Special Province); inner suburbs = *kotas* (municipalities) bordering Jakarta; outer suburbs = *kabupatens* (districts) bordering Jakarta; inner suburbs + outer suburbs = Bodetabek; CBD + Bodetabek = Jabodetabek (Jakarta Metropolitan Area or JMA); extra outer suburbs + JMA = Greater Jakarta.

Two other highways, Jakarta-Cikampek Toll Road and Jakarta-Merak Toll Road, were built along the northern coastal zone of Java. Most of the industrial parks or planned industrial concentration areas built since 1989 are located along these later coastal highways. With a length of about 90 km each, the last two highways extended the traditional boundaries of the metropolitan area, forming the western and the eastern *extra* outer suburbs, respectively (see Fig. 1).

With the opening of the last two major highways, it is no longer possible to apprehend Jakarta's industrial suburbanization by studying Jabodetabek alone. The industrial parks have in the past two decades expanded beyond the boundaries of Jabodetabek toward Serang District in the west and Karawang District in the east (see Fig. 2). This extended mega-city region (Greater Jakarta) covers a total area of 9,016.43 km<sup>2</sup>. There are now more than 35 industrial parks in Greater Jakarta with a total area of over 18,000 hectares. With another property boom in the offing for Indonesia, it is likely that their number will rise considerably in the coming years in order to accommodate a massive influx of FDI relocations from other countries impacted by the last global recession (CNTV, 2012).

The size of industrial parks in Greater Jakarta ranges from 50 to 1,800 hectares, while the average size is about 500 hectares. These industrial parks tend to cluster in several locations. At least six suburban towns and sub-districts, hereafter referred to as industrial centers, have industrial areas of more than the average size of 500 hectares.

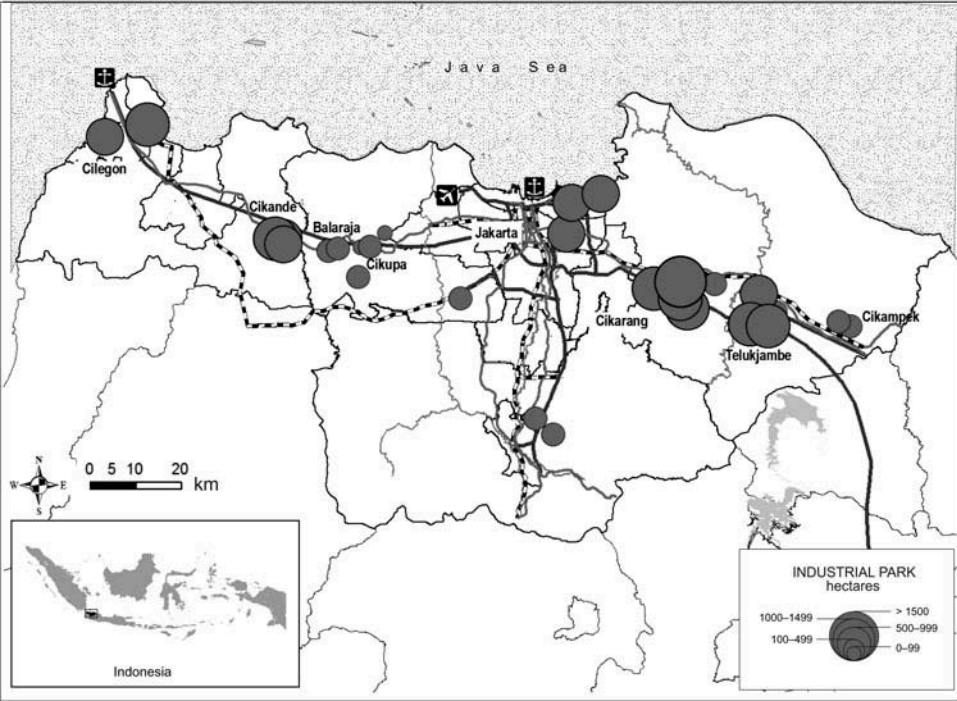


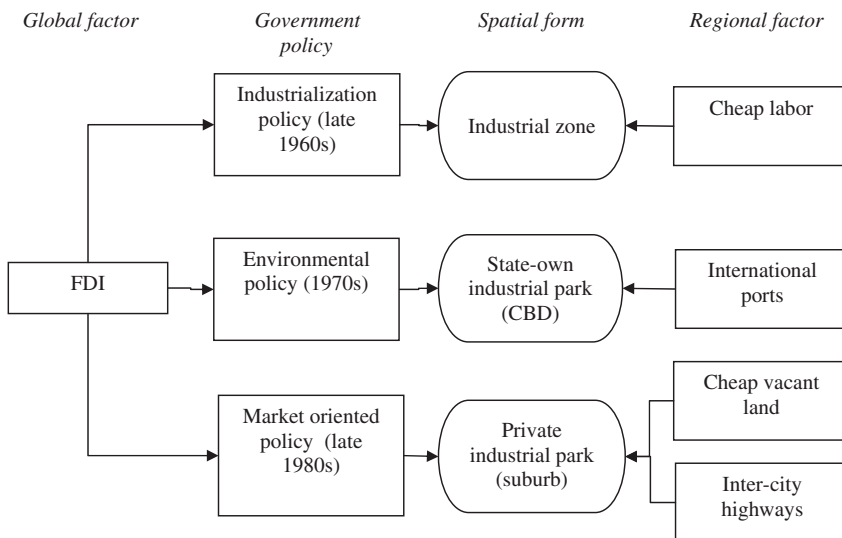
Fig. 2. Map of major industrial parks in Greater Jakarta.

Major industrial centers in Bodetabek (the outer suburbs) include Cikupa-Balaraja (Tangerang District) and Cikarang (Bekasi District). In fact Cikarang, with a total industrial land area of nearly 6,000 hectares, has in the past two decades become the largest planned industrial center in Southeast Asia (Hudalah and Firman, 2012). Four other industrial centers are located beyond Jabodetabek in the extra outer suburbs, which include Serang (Cilegon and Cikande) and Karawang (Telukjambe and Cikampek).

#### *Major Factors and Actors Triggering Industrial Deconcentration*

Suburbanization of economic activity in Greater Jakarta was triggered in the 1980s by, amongst other things, the national government's deregulation and debureaucratization measures. These market-oriented policies aimed to attract domestic and foreign private investors in finance and manufacturing industries in order to accelerate economic growth. The policies not only boosted FDI in manufacturing, but also real estate development, leading to uncontrolled growth of large-scale projects by private land developers in the suburbs (Firman, 2000). In addition to the influence of this government-supported global capitalism, other institutional factors contributing to Greater Jakarta's suburbanization include the rise of a middle-class society, clientelist governance practices, and the weakened presence of government in the suburbs (Hudalah and Woltjer, 2007; Hudalah et al., 2007).

Industrial development has played an important role in Jakarta's suburban transformation (see Fig. 3). Government policies with respect to industrial investment in fact date back to the beginning of Suharto's New Order Regime in the late 1960s. With its market-led development policy, the New Order administration positioned industrialization as a



**Fig. 3.** Conceptual diagram of industrial deconcentration in Greater Jakarta.

key policy platform to boost economic growth. Industrial development was first endorsed in industrial zones (*zona industri*). An industrial zone refers to a concentration of industrial activities that are not necessarily supported by planned and adequate infrastructure and facilities. Traditionally, local and labor-intensive industries tended to locate in such unplanned industrial areas.

Meanwhile, foreign and hi-tech industrial investors preferred to operate in industrial parks (*kawasan industri*). Kwanda (2000) found that in the early stages of their development in the 1970s, industrial parks were mainly built by government enterprises as a reaction to the increasing environmental impact and infrastructure inadequacy of the industrial zones. Most of these government-supported industrial parks were developed in the northern/coastal zone of the inner-city of Jakarta. Among others, the Jakarta Industrial Park Pulogadung (594 ha) and Kawasan Berikat Nusantara (595 ha), built in 1970 and 1986 respectively, remain in operation. Later, the private sector became the main actor in the development of suburban industrial parks, the first of which was first built in Tangerang District (Cikupa-Balaraja).

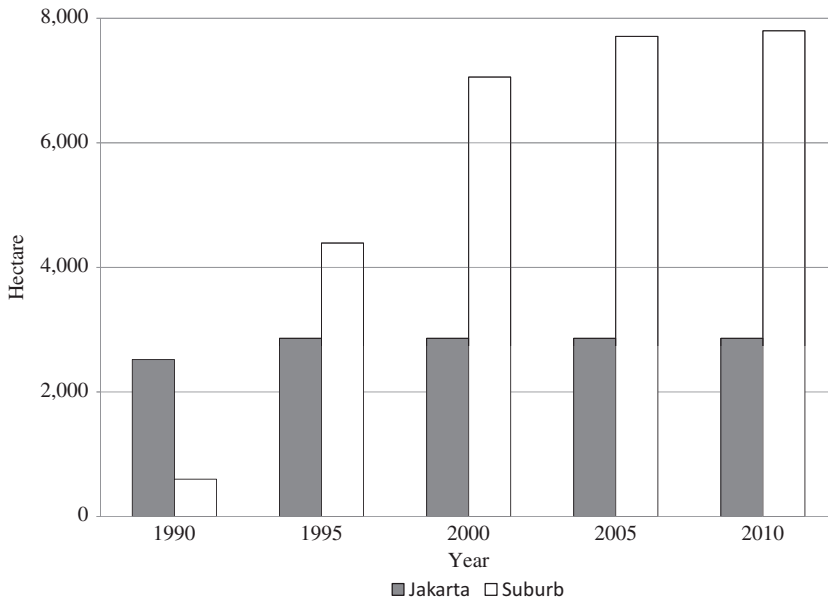
In response to a growing number of proposals by domestic as well as foreign private investors, through Presidential Decree 53 (1989), the government further facilitated the private sector's participation in large-scale industrial-park development in Indonesia. Another presidential decree was later issued to set guidelines for industrial-park development. Presidential Decree 41 (1996) introduced industrial parks as centers for industrial activities with the provisioning of infrastructure and supporting facilities developed and operated by licensed industrial-park companies. Industrial-park development focused on accelerating industrialization of the regions, facilitating industrial activities, directing industrial location, and strengthening environment-friendly industrialization.

These presidential decrees have encouraged new foreign investors to build plants in industrial parks. As a result, in 2011, BKPM (the Investment Coordinating Board) granted 317 new FDIs permanent licenses relating to manufacturing industries in Jabodebek (JMA minus Tangerang)—compared to 75 domestic investments. These FDIs originate from various countries. In Jababeka Industrial Estate alone, for example, there are more than 1,500 national and multinational companies from more than 30 countries, including the United States, Britain, France, Germany, the Netherlands, Australia, Japan, South Korea, China, Taiwan, etc. (PT Jababeka, 2010).

Since the end of the 1980s, no new industrial parks have been developed in the city. Available industrial land in Jakarta has declined. In contrast, during the 1990s, after the enactment of the presidential decrees, massive private industrial-park development took place in the suburbs. The suburban industrial parks were mostly concentrated in Cikarang (Bekasi District).

Until 1994, 146 private industrial-park locations were registered in HKI (*Himpunan Kawasan Industri Indonesia*) or the Indonesia Industrial Park Association. Most of these registered industrial parks are located in the suburbs of Jakarta (Dura-Guimera, 2003). Since the middle of the 1990s, the suburbs have taken over the role of the city in leading the supply of formal industrial land (see Fig. 4).

In addition to pro-market government policy and pressure from global capital, suburban industrial-park development can also be regarded as an inevitable consequence of an increasingly congested Jakarta. Since the 1990s, it has gradually become more difficult



**Fig. 4.** Industrial land supply in Greater Jakarta.

Source: Analyzed from Colliers International (1997, 2002, 2007, 2010).

to find vacant land parcels in the city to accommodate new industries. As a result, during the first property boom—until its peak in 1997—the industrial land price in the city was about three times higher than that in the inner and outer suburbs (Bodetabek). Since the mid-2000s, the city has no longer been able to supply land for new large-scale industries (see Table 1).

The overall industrial land price in the whole metropolitan area fell substantially following the Asian economic crisis in 1998. However, since the second half of the

**TABLE 1.** LAND-USE PRICE MEDIAN IN GREATER JAKARTA'S MAJOR INDUSTRIAL PARKS (\$/M<sup>2</sup>)

| City/district   | 1997  | 2002 | 2007         | 2011         |
|-----------------|-------|------|--------------|--------------|
| Jakarta         | 228.5 | 77.5 | <sup>a</sup> | <sup>a</sup> |
| Suburb          | 74.5  | 42.7 | 61.0         | 116.7        |
| Inner and outer | 92.0  | —    | 66.2         | 123.9        |
| Tangerang       | —     | —    | 85.0         | 113.3        |
| Bekasi          | 91.5  | —    | 58.5         | 150.0        |
| Bogor           | —     | —    | 55.0         | 108.3        |
| Extra outer     | 63.8  | —    | 38.5         | 91.7         |
| Karawang        | 70.0  | —    | 40.0         | 100.0        |
| Serang          | 57.5  | —    | 37.0         | 83.3         |

<sup>a</sup>No new large-scale land available.

Source: Calculated from Colliers International (1997, 2002, 2007, 2010).

2000s, land prices have started to experience positive growth again. In fact, they are now peaking again due to the expected large-scale influx of FDI relocation from countries affected by the recession that began with the financial crisis in 2008. In the coming years, pressure will probably ease as new industrial parks and towns are now being constructed in the extra outer suburbs of Karawang and Serang.

While gradually losing its significance, Jakarta, where the international seaport and airport are currently located, is still the main hub for the national and global markets of industrial products. Therefore, as indicated in Table 1, during the last decade, spatial proximity to this metropolitan hub has still impacted on the spatial distribution of industrial land-use prices. However, the spatial distribution of land prices will potentially shift as new airports and a seaport on a scale comparable to that of Jakarta are currently being planned in the eastern extra outer suburbs (Karawang and Majalengka) (Kementerian Koordinator Bidang Perekonomian, 2011).

THE DYNAMICS OF MANUFACTURING EMPLOYMENT IN GREATER JAKARTA

This section describes the shifting spatial distribution of manufacturing employment in Greater Jakarta in 1995–2010. Figure 5 depicts the spatial distribution of manufacturing employment in Greater Jakarta based on the E/P ratio in 1995. This first reference point represents the period when the policies and regulations facilitating private involvement in

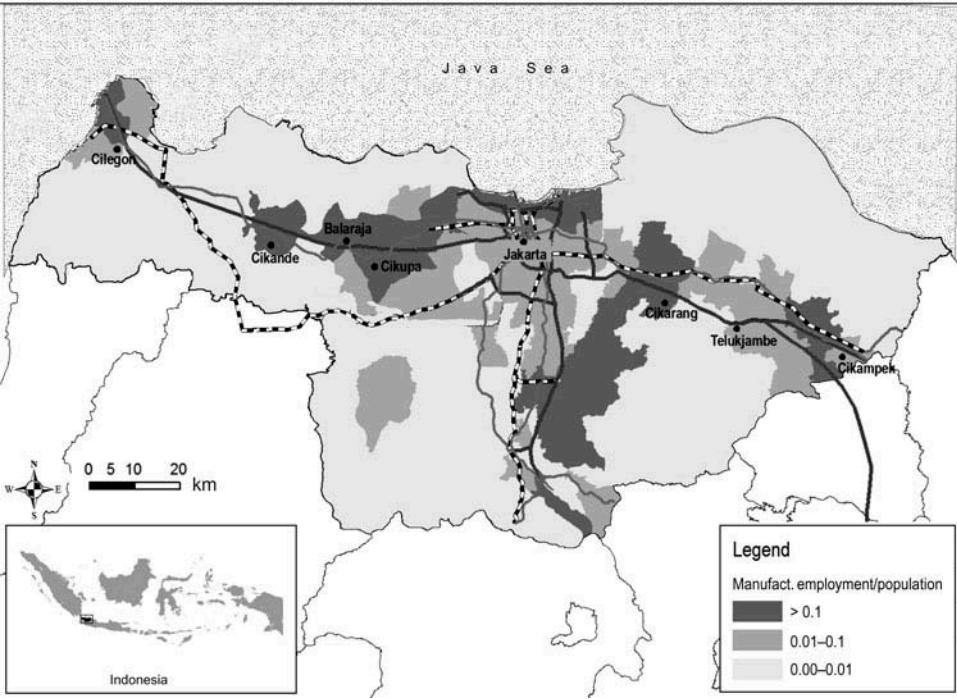


Fig. 5. Manufacturing employment-population ratio in Greater Jakarta 1995.

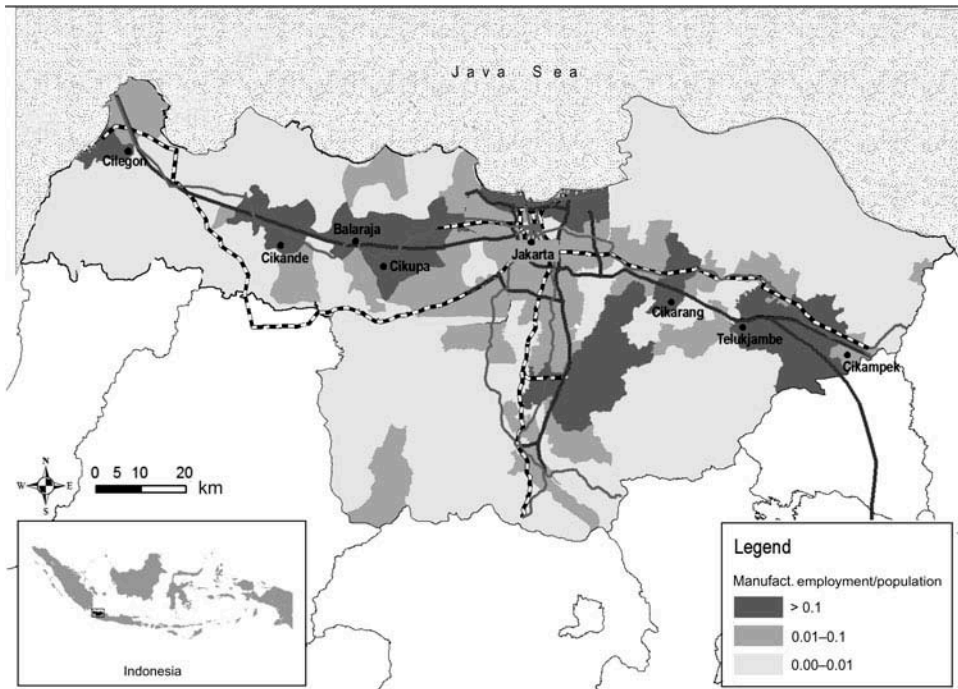


Fig. 6. Manufacturing employment-population ratio in Greater Jakarta 2001.

industrial and land development had been implemented and were beginning to take effect. The figure shows that even in this early phase of formal industrial land development, employment zones (referring to subdistricts with  $E/P > 0.1$ ) were dispersed across the metropolitan area. Within Jakarta, employment zones can only be found in the northern part. Meanwhile, a considerable number of new employment zones are present in the suburbs. The total area of these suburban employment zones is much larger than of those in Jakarta.

In 2001, the  $E/P$  ratios show an extension of the manufacturing dispersal already noted in 1995. First, there is a decrease of manufacturing employment in Jakarta. Meanwhile, a different trend can be found in the suburbs where the location of employment zones has shifted considerably. As indicated in Fig. 6, employment concentration has declined in the southern suburbs, where unplanned industrial zones have long since agglomerated in Cibinong, Bogor District. Meanwhile, the eastern suburbs have received a significant influx of employment, filling in the planned industrial centers along the highway. As another important trend, employment zones are seen leaving the inner suburbs (the municipalities adjacent to Jakarta). They have been pushed further toward the eastern suburban districts, especially in the planned industrial centers of Cikarang (Bekasi District) and Telukjambe (Karawang District). At a smaller scale, such a locational dynamic can also be identified in the western extra outer suburbs, notably Cilegon, where employment concentration has shifted from an unplanned industrial zone in the northern part of the city to a planned industrial center located in the south (Ciwandan).

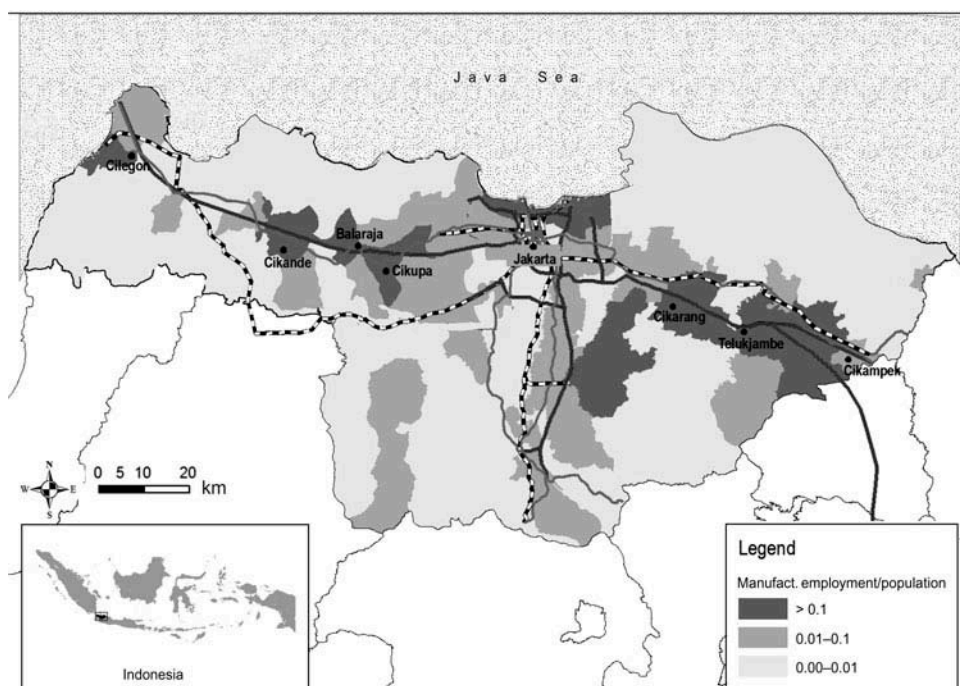


Fig. 7. Manufacturing employment-population ratio in Greater Jakarta 2010.

The most recent spatial distributions of E/P ratios are illustrated in Fig. 7. If we compare this latest figure with the earlier ones, we can see that, notwithstanding the smaller shifts already pointed out, in the past 15 years there has been no significant change of major employment centers in central Jakarta. This relatively static situation implies that manufacturing employment in the center has reached its saturation point.

In comparison, employment centers in the suburbs were relatively dynamic. First, the 1990s–2000s reveal a trend of locational change of suburban manufacturing mainly from industrial zones to industrial parks. Second, the 2000s–2010s reveal a process of manufacturing intensification in existing suburban industrial parks around Jakarta. The latter is evident from the decline in the number and total area of suburban industrial centers in Greater Jakarta that show positive growth in manufacturing employment.

These dynamics of manufacturing employment between Jakarta and the suburbs are shown in Table 2. In 1995 Jakarta (the CBD) accounted for only 36% of total metropolitan employment. Furthermore, the manufacturing employment share of Jakarta decreased by 15% between 1995 and 2010. This figure strengthens the finding of an earlier study that reports that since the beginning of the 1990s most formal manufacturing employment has been located outside Jakarta (Henderson and Kuncoro, 1996). This deindustrialization extends toward the inner suburbs, namely municipalities directly bordering central Jakarta. These municipalities include Tangerang, Bekasi, and South Tangerang Municipalities. This

**TABLE 2.** MANUFACTURING EMPLOYMENT IN GREATER JAKARTA (WORKER)

| Metropolitan zone              | Employment |           | Growth   |            |
|--------------------------------|------------|-----------|----------|------------|
|                                | 1995       | 2010      | Absolute | Percentage |
| CBD (Jakarta Capital Province) | 376,386    | 321,270   | -55,116  | -15        |
| Bodetabek                      |            |           |          |            |
| Inner suburbs                  | 278,354    | 256,451   | -21,903  | -8         |
| Outer suburbs                  | 323,072    | 446,199   | 123,127  | 38         |
| Extra outer suburbs            | 78,587     | 203,789   | 125,202  | 159        |
| Total (Greater Jakarta)        | 1,056,399  | 1,227,709 | 171,310  | 16         |

implies that the CBD (Jakarta City) and its adjoining municipalities have lost their dominant role as the long-established manufacturing center of the metropolitan area.

In contrast, the outer suburbs (Bogor, Tangerang and Bekasi Districts) have experienced a rapid influx of manufacturing employment. The most dramatic growth has occurred in the extra outer suburbs, where employment has increased by 159% during the 15-year period alone. This positive relationship between employment increase and distance to the CBD provides a strong indication that formal manufacturing in Greater Jakarta has extended and has increasingly suburbanized.

#### *Scatteration or Polycentricity?*

Based on the industrial-park data compilation and E/P analysis, at least three spatial forms of manufacturing suburbanization are expected to emerge in Greater Jakarta: (1) non-concentration zones, (2) unplanned concentration zones, and (3) planned concentration zones. We use  $E/P = 0.1$  as the threshold distinguishing non-concentration from concentration zones. A sub-district or town is called a manufacturing concentration zone if its E/P in 1995, 2001, or 2010 is above 0.1. Concentration zones can be planned or unplanned. An unplanned concentration zone refers to places dominated by industrial zones. Meanwhile, a planned concentration zone is where most industrial parks are clustered. Six out of the seven industrial centers identified earlier meet the criteria for this latter category: North Jakarta (the metropolitan core), Cikarang (Bekasi), Telukjambe (Karawang), Cikupa-Balaraja (Tangerang), Cikande (Serang), and Ciwandan (Cilegon). Cikampek (Karawang) is not included because this suburban industrial center has an  $E/P < 0.1$ . The last spatial form resulting from the analysis are non-concentration zones, which include the remainder of the sub-districts scattered throughout the metropolitan area with relatively low employment density, indicated by  $E/P \leq 0.1$ .

In 1995, employment in the unplanned concentration zones represented about one-third of the one million metropolitan manufacturing jobs in Greater Jakarta. This share was, however, been reduced by 6% by 2010. A similar trend has been experienced by the non-concentration zones, whose share dropped from 36% to 31% over the period. Together the unplanned concentration and non-concentration zones had an employment share of 72%

**TABLE 3.** THE SPATIAL FORMS OF MANUFACTURING EMPLOYMENT IN GREATER JAKARTA

| Spatial form            | Employment |           | Growth   |            |
|-------------------------|------------|-----------|----------|------------|
|                         | 1995       | 2010      | Absolute | Percentage |
| Planned concentration   | 298,749    | 496,672   | 197,923  | 66         |
| Unplanned concentration | 375,982    | 351,633   | -24,349  | -6         |
| Non-concentration       | 381,668    | 379,404   | -2,264   | -1         |
| Total (Greater Jakarta) | 1,056,399  | 1,227,709 | 171,310  | 16         |

in 1995 but 15 years later this had fallen to 60%. This proportion, whilst declining, remains high, and suggests that most manufacturing employment is still scattered across the metropolitan area. However, the negative trend implies that this scattered pattern has been decreasing considerably.

Indeed, over the 15-year period under study, there has been a shift in manufacturing employment from unplanned and sprawling zones toward planned concentration in sub-urban industrial centers. Initially, in 1995 the industrial centers attracted nearly 300,000 jobs (28% of the total). However, as presented in Table 3 the number rose by 66% in 2010 to almost 500,000 (40% of the total). In view of this trend it can be suggested that, if they continue, a polycentric spatial structure will emerge in the region. The numbers also suggest that the private sector—which controls most of the planned industrial centers in the suburbs—is playing a substantial role in realizing the expected new metropolitan structure.

#### *Specialization of Industrial Centers in Greater Jakarta*

Each of the employment areas identified has its own industrial structure, which we explore in this section. Using Location Quotient (LQ) analysis, we first identify whether the industrial centers are specialized (and, if so, in what sectors). We then track shifts in sectoral specialization among the industrial centers between 1995 and 2010.

The classification of business sectors used by the Central Statistical Bureau in 2010 is more complex than that of 1995. In order to make the longitudinal comparison possible, we first synchronized the classification by using *Table of Business Sector Matching 1990–2000* (Badan Pusat Statistik, 2000). As the result, the manufacturing sectors used in the LQ analysis of 2010 consists of 16 sectors: food, textiles, clothing, leather, woods and household equipment, papers, publishing, chemicals, petroleum, coal, rubber and plastics, mining, metals, machinery, electronics, automotive, and other manufacturing. Due to data limitation, in the LQ analysis of 1995, several sectors were merged and simplified to nine groups of business sector: metals; energy (coal and petroleum), chemical, and rubber and plastics; woods and household equipment; paper and publishing; mining; foods; textiles, leather, and clothing; machinery, electronics, and automotive and; other manufacturing.

**TABLE 4.** LQ FOR INDUSTRIAL CENTERS IN 1995

| Industrial sector                          | Cilegon            | Cikande | Cikupa-Balaraja | Cibitung | CBD (Jakarta) |
|--------------------------------------------|--------------------|---------|-----------------|----------|---------------|
| Metals                                     | 18.87 <sup>a</sup> | 0.00    | 0.44            | 3.38     | 0.64          |
| Energy, chemicals, and rubber and plastics | 1.61               | 2.82    | 1.64            | 0.82     | 0.61          |
| Wood                                       | 0.00               | 1.40    | 1.93            | 0.98     | 0.78          |
| Paper and publishing                       | 0.00               | 2.42    | 1.34            | 1.39     | 0.88          |
| Mining                                     | 0.67               | 0.00    | 0.97            | 2.70     | 0.71          |
| Foods                                      | 0.00               | 0.00    | 1.17            | 1.01     | 1.16          |
| Textile, leather, and clothing             | 0.00               | 0.48    | 0.64            | 1.11     | 1.19          |
| Machinery, electronics, and automotive     | 2.15               | 0.22    | 0.75            | 0.06     | 1.14          |
| Other                                      | 0.00               | 2.20    | 1.40            | 2.09     | 0.83          |

<sup>a</sup>The extremely high LQ is due to small employment share of Cilegon.

Meanwhile, the industrial centers included in this analysis are those with  $E/P > 0.1$ . In 1995, there were five industrial centers that met the criteria, which are North Jakarta (part of Jakarta or the CBD of Greater Jakarta) and four other industrial centers in the suburbs: Cikupa-Balaraja, Cikande, Cilegon, and Cikarang. Using the same criteria, Telukjambe industrial center in Karawang District was later added in the LQ analysis for the year 2010. During this 15-year period, the total employment in the selected industrial centers doubled from about 280,000 to nearly 550,000 jobs.

In 1995, with a total of about 170,000 jobs, the CBD still played a role as the industrial powerhouse of the metropolitan area. It was the primary location for a large number of manufacturing sectors including foods, textile, leather, clothing, machinery, electronics, and automotive (Table 4). However, this shifted considerably over the subsequent 15 years. As the number of jobs declined to 160,000, in 2010 the CBD was only able to maintain its leading role in consumer goods sectors such as foods, clothing, and publishing (Table 5). From an employment specialization perspective, in the past decades the CBD's manufacturing domination over the metropolitan area has gradually diminished. Most manufacturing sectors whose operations require relatively large plots of land or are highly dependent on natural resources (i.e., water) such as textile, leather, machinery, electronics, and automotive have in the past two decades gradually relocated to the suburbs.

In line with the relative decline of Greater Jakarta's CBD, in 1995 specialization and diversification of manufacturing sectors had already started to occur in the suburbs (Table 4). The outer suburbs, which at that time consisted of Cibitung, Cikupa-Balaraja, and Cikande, served as the major location for polluting and labor-intensive industries such as mining, wood, and household equipment. Meanwhile, the extra outer suburban industrial center or Cilegon focused on accommodating heavy and extractive industries, especially metals and chemicals, including petroleum and coal mining and their derivatives.

The functional separation between the CBD, the inner suburbs, the outer suburbs, and the extra outer suburbs becomes more obvious by 2010 (Table 5). In fact, there is a connection between the development of industrial parks and the specialization of

**TABLE 5.** LQ FOR INDUSTRIAL CENTERS IN 2010

| Industrial sector   | Cilegon            | Cikande | Cikupa-Balaraja | Karawang | Cikarang | CBD (Jakarta) |
|---------------------|--------------------|---------|-----------------|----------|----------|---------------|
| Wood                | 7.99               | 0.00    | 1.95            | 0.05     | 0.57     | 0.85          |
| Energy              | 15.30 <sup>a</sup> | 0.81    | 0.52            | 1.71     | 0.46     | 0.74          |
| Chemicals           | 6.07               | 0.58    | 0.66            | 0.12     | 1.07     | 1.13          |
| Metals              | 3.68               | 0.73    | 1.17            | 0.99     | 1.21     | 0.50          |
| Leather             | 0.02               | 6.27    | 2.30            | 0.10     | 0.32     | 0.10          |
| Textile             | 0.00               | 1.36    | 1.68            | 1.35     | 0.90     | 0.55          |
| Paper               | 0.00               | 0.00    | 1.11            | 2.57     | 1.46     | 0.23          |
| Mining              | 0.56               | 0.37    | 1.62            | 0.61     | 1.31     | 0.52          |
| Rubber and plastics | 0.24               | 0.26    | 0.94            | 1.53     | 1.33     | 0.72          |
| Machinery           | 0.77               | 0.08    | 0.81            | 2.12     | 1.70     | 0.20          |
| Electronics         | 0.00               | 0.12    | 0.04            | 3.49     | 2.01     | 0.01          |
| Automotive          | 0.00               | 0.03    | 0.40            | 0.90     | 1.11     | 1.60          |
| Foods               | 2.80               | 1.22    | 0.95            | 0.35     | 0.67     | 1.42          |
| Clothing            | 0.00               | 0.17    | 0.76            | 0.08     | 0.18     | 2.63          |
| Publishing          | 0.00               | 0.16    | 0.59            | 0.00     | 0.68     | 2.19          |
| Other               | 0.00               | 0.11    | 1.06            | 0.27     | 1.13     | 1.30          |

<sup>a</sup>The extremely high LQ is due to small employment share of Cilegon.

manufacturing deconcentration. First, an increasingly large number of relatively high-tech industries (in the form of machinery, electronics, and automotive) have moved out of the CBD, but only toward the newly emerging, larger-scale, and planned industrial parks. Most of these industrial parks are located in the eastern outer suburbs, notably Cikarang and Karawang. In comparison, the older, smaller-scale, and more fragmented industrial centers have historically been concentrated in the western counterparts, namely Cikupa-Balaraja and Cikande. Most of them were built before the period of massive enactment of pro-market industrial and investment policies of the late 1980s. By 2010, they had received the largest influx of low-cost labor and polluting industries such as textile and leather, which previously were also considered a primary sector associated with the CBD. Finally, Cilegon is the only industrial center of the extra outer suburbs with an E/P ratio above 0.1. It continues to play the role of center for heavy and polluting industries. In addition, a considerable number of wood, household equipment and other extractive industries have also recently shifted from the outer suburbs toward this extra outer suburban center.

## CONCLUSION

It has been suggested that studies of intra-metropolitan industrial deconcentration are particularly important in order to understand the spatial and socio-economic dynamics of global cities in industrializing countries (Feng and Zhou, 2005; Feng et al., 2008; Wu and

Phelps, 2008; Zhou and Ma, 2000). Greater Jakarta has been a fast-growing global city-region in Southeast Asia, and has witnessed considerable industrial deconcentration. Since the early 1990s most manufacturing employment has been being suburbanized. Industrial deconcentration has now in fact reached an advanced phase. This deconcentration is indicated not only by the declining role of the CBD but also by the extended boundaries of the metropolitan area.

In Indonesia, the role of the government in the process of large-scale industrial deconcentration has been relatively modest, focused on the facilitation of private sector involvement in industrial investment and development. It is evident that the private sector has played a substantial role in accompanying the deconcentration of formal manufacturing employment. In contrast, there is no indication that spatial planning has played any crucial role in guiding this deconcentration. As reported elsewhere, metropolitan spatial planning in Indonesia tends to be fragmented and lacks the institutional capacity to anticipate rapid suburbanization (Henderson and Kuncoro, 1996; Hudalah et al., 2007).

A large proportion of formal manufacturing employment in Greater Jakarta is still located in unplanned corridors of industrial zones along major inter-city highways. Another portion of industrial employment is still scattered across the suburbs. However, this tendency toward unplanned industrial development has diminished considerably. Partly due to stricter environmental regulation, a significant number of industries have been leaving the unplanned suburban zones and scattered pockets, filling out industrial centers where most of the private industrial parks agglomerate. These suburban industrial centers have now diversified, taking over major manufacturing employment sectors previously located in the CBD. These private industrial centers appear to be altering the spatial structure of the metropolitan area, from a scattered toward a relatively more polycentric spatial arrangement.

This changing spatial structure can be seen as an opportunity to promote a better planned metropolitan area. However, with an absence of adequate planning framework and institutions, the active roles of foreign investors and big private developers in suburban industrial transformation may potentially strengthen social segregation and infrastructure fragmentation and reduce the quality of the physical environment (Hudalah and Firman, 2012; Hudalah et al., 2007). Therefore, there is an urgent need to improve the capacity of suburban local government and metropolitan governance in order to better formulate, adapt, coordinate, and implement local and regional land-use plans and integrate them with regional economic policies and longer-term plans prepared by the national government.

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